# THICK GENERATION OF DRINFELD–KOHNO CATEGORIES FOR ALL SIMPLE LIE TYPES VIA MULTIPLICITY-FREE $\ell$-WEIGHTS

RAEEZ LORGAT

Abstract. For every finite-dimensional simple Lie algebra $\mathfrak{g}$, we prove that the Drinfeld–Kohno category $\mathrm{DK}_{\mathfrak{g}}$ is thickly generated by its fundamental evaluation representations, replacing the minuscule hypothesis of Kazhdan–Lusztig and earlier type-$A$ work by the strictly weaker multiplicity-free $\ell$-weight property of Chari–Moura. The proof proceeds in four steps: a character identity, transfer of the multiplicity-free $\ell$-weight property from quantum affine algebras via Gautam–Toledano Laredo, Hernández block separation at generic spectral parameters, and a character accounting argument inside category $\mathcal{O}^{\mathrm{sh}}$. As a corollary, the degree-3 component of the Maurer–Cartan element of the chiral algebra determines all higher degrees for every simple Lie type, including the exceptional algebra $E_8$, which admits no minuscule representation but does admit multiplicity-free $\ell$-weights.

## 1. Introduction

1.1. **The minuscule hypothesis.** The classical Drinfeld–Kohno theorem [20, 5] identifies the monodromy of the Knizhnik–Zamolodchikov connection on conformal blocks for $\widehat{\mathfrak{g}}_k$ with the $R$-matrix action of the quantum group $U_q(\mathfrak{g})$ at $q = e^{\pi i/(k+h^\vee)}$. Kazhdan and Lusztig extended this to a braided tensor equivalence between the Kazhdan–Lusztig category at level $k$ and the category of type-1 modules over the quantum group [16, 17, 18, 19]. Their proof, and the subsequent refinement of Etingof and Kazhdan in terms of quantization of Lie bialgebras [6, 7], leans at several points on the availability of a *minuscule* representation: a fundamental representation whose nonzero weight spaces are all one-dimensional and form a single Weyl group orbit.

The minuscule hypothesis organizes the block structure of the Kazhdan–Lusztig category and, in the Yangian or quantum affine setting, controls the decomposition of tensor products with prefundamental modules. It is an *attractive* hypothesis because type $A$ is profligate in minuscule representations:

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every fundamental representation $V_{\omega_i}$ of $\mathfrak{sl}_n$ is minuscule, so the minuscule hypothesis is automatic. It is a *constraining* hypothesis because outside type $A$ minuscule representations are rare and, in the exceptional algebras $F_4$, $G_2$, and most critically $E_8$, nonexistent.

The absence of a minuscule representation in $E_8$ is not a technicality but a structural feature of its root system: the adjoint representation of $E_8$ is simultaneously its smallest nontrivial and its smallest fundamental representation, the weight spaces of nonzero roots are all one-dimensional but the zero weight space has dimension $\mathrm{rk}(E_8) = 8$, and no other Weyl group orbit of weights exhausts the nonzero weight spaces of a fundamental representation. The standard literature therefore omits $E_8$ from the statement of the categorical Drinfeld–Kohno theorem, or formulates it in a way that requires separate proof.

1.2. **Multiplicity-free $\ell$-weights.** The correct replacement for the minuscule hypothesis was discovered in a very different context: the theory of $q$-characters of quantum affine algebras, due to Frenkel and Reshetikhin [10] and extended by Frenkel–Mukhin [11], Nakajima [22], and Chari–Moura [1]. To each finite-dimensional simple representation $L(\Psi)$ of a quantum affine algebra $U_q(\hat{\mathfrak{g}})$ one associates a *Drinfeld polynomial* $\Psi$ and a *$q$-character* $\chi_q(L(\Psi))$, a Laurent polynomial in formal variables $Y_{i,a}$ whose monomials record the $\ell$-weights: the joint eigenvalues of a commuting family of quantum loop generators $\phi^{\pm}_{i,\pm}(z)$.

A representation is called *multiplicity-free in $\ell$-weights* if every monomial of its $q$-character appears with coefficient 1. This is a stronger condition than multiplicity-freeness of classical weights (on which the minuscule hypothesis rests) because the loop generators split classical weight spaces into finer pieces, and at generic spectral parameters these pieces are in fact one-dimensional for fundamental representations of every simple Lie type.

**Theorem 1.1** (Chari–Moura; [1, 22, 11])**.** *For every finite-dimensional simple Lie algebra $\mathfrak{g}$ and every fundamental weight $\omega_i$, the fundamental evaluation module $V_{\omega_i}(a)$ of $U_q(\hat{\mathfrak{g}})$ has multiplicity-free $\ell$-weights: its $q$-character has exactly* $\dim V_{\omega_i}$ *distinct monomials.*

This is the input that powers the present paper. The fundamental representations of $U_q(\hat{\mathfrak{g}})$ are special (in the Frenkel–Mukhin sense), so the Frenkel–Mukhin algorithm computes their $q$-characters uniformly in type; Nakajima's geometric computation confirms the simply-laced cases; Chari–Moura give explicit closed-form formulas for the classical types; and the exceptional types $E_6, E_7, E_8, F_4, G_2$ are covered by the algorithm whose output is multiplicity-free by direct inspection.

1.3. **Statement of the main theorem.** Let $Y_\hbar(\mathfrak{g})$ denote the Yangian of a simple Lie algebra $\mathfrak{g}$ with formal parameter $\hbar$. For a fundamental weight $\omega_i$ and spectral parameter $a \in \mathbb{C}$, let $V_{\omega_i}(a)$ be the corresponding fundamental evaluation module, and let $L_i^-(b) \in \mathcal{O}^{\mathrm{sh}}$ denote the negative prefundamental

simple in Hernández' shifted category, of highest $\ell$-weight indexed by the Drinfeld polynomial of the identity representation with spectral shift $b$ (see Definition 2.3).

**Theorem 1.2** (Main theorem)**.** *For every finite-dimensional simple Lie algebra $\mathfrak{g}$, every fundamental weight $\omega_i$, and all generic spectral parameters $a, b \in \mathbb{C}$, there is a direct sum decomposition of $Y_\hbar(\mathfrak{g})$-modules in $\mathcal{O}^{\mathrm{sh}}$:*

$$V_{\omega_i}(a) \otimes L_i^-(b) \;\cong\; \bigoplus_{m \in \chi_q(V_{\omega_i}(a))} L_i^-(b \cdot m). \tag{1.1}$$

*The direct sum has exactly $\dim V_{\omega_i}$ summands, each appearing with multiplicity one; the shift factor $m$ runs over the monomials of the $q$-character of $V_{\omega_i}(a)$. In particular, the Drinfeld–Kohno category $\mathrm{DK}_{\mathfrak{g}}$ of $Y_\hbar(\mathfrak{g})$-modules in $\mathcal{O}^{\mathrm{sh}}$ is thickly generated by the fundamental evaluation objects together with the negative prefundamental objects.*

The restriction to generic spectral parameters is genuinely generic: it excludes only a finite set of values of $a-b$ at which Hernández $A$-factorizations occur, at which (1.1) is replaced by a filtration with the same associated graded.

Theorem 1.2 replaces the minuscule hypothesis of Kazhdan–Lusztig and of earlier type-$A$ treatments of the categorical Clebsch–Gordan decomposition (see [2, §11] and the Francis–Gaitsgory pro-nilpotent completion of factorization categories [9]). The key observation is that the minuscule hypothesis was never the *ontological* reason for the decomposition: it was a *computational* shortcut that forced classical weight multiplicities to be trivial. The correct hypothesis is the multiplicity-freeness of $\ell$-weights, which distinguishes the fine structure that the loop generators see.

1.4. **Consequence: MC3 for all simple types.** The chiral bar complex $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ of a vertex algebra $\mathcal{A}$ admits a Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ whose degree-2 projection is the classical $r$-matrix and whose higher-degree projections form the shadow tower; see [21, Vol. I]. The third master conjecture (MC3) of the modular Koszul duality programme asserts that the degree-3 component of $\Theta_{\mathcal{A}}$, when restricted to the evaluation-generated core, determines all higher-degree components through thick generation and pro-nilpotent completion of the ambient factorization category. The present paper removes the only remaining Lie-type obstacle to MC3 for Yangian targets:

**Corollary 1.3** (MC3 for all simple types)**.** *For every finite-dimensional simple Lie algebra $\mathfrak{g}$, the categorical Clebsch–Gordan step of MC3 for the Yangian $Y_\hbar(\mathfrak{g})$-target holds: the degree-3 shadow projection $\pi_{3,0}(\Theta_{\mathcal{A}}^{E_1})$, combined with the evaluation-core Drinfeld–Kohno comparison, determines all higher-degree projections of the Maurer–Cartan element on the thick subcategory generated by the fundamental evaluation modules and the prefundamental simples.*

1.5. **Relation to earlier work.** Kazhdan and Lusztig [16] treat the equivalence with quantum group modules type by type, with the minuscule weights of the classical types doing the combinatorial work. Etingof and Kazhdan [6, 7] replace this by a universal quantization functor on Lie bialgebras; their proof is type-uniform, but the passage to the Yangian or quantum affine setting reintroduces the minuscule hypothesis through the prefundamental Clebsch–Gordan decomposition of Chari–Pressley [2]. Hernández and Jimbo [13] introduced the shifted category $\mathcal{O}^{\mathrm{sh}}$ and proved a block criterion which is the combinatorial engine of our Step 3. Francis and Gaitsgory [9] developed the pro-nilpotent completion formalism that allows us to pass from a comparison on the compact core to the full shifted category.

The new content is neither a new quantum group identity nor a new factorization formalism, but the recognition that Theorem 1.1 is the correct replacement for the minuscule hypothesis in the proof of the categorical Clebsch–Gordan decomposition. In the language of the Lorgat monograph [21], we resolve the categorical CG package of MC3 for every simple Lie type, including $E_8$, in which no minuscule representation exists.

1.6. **Organization.** Section 2 recalls Lie bialgebras, the Drinfeld–Kohno category, the chromatic filtration on compact objects, and the formal definition of $\ell$-weights and the multiplicity-free property. Section 3 states the main theorem precisely and establishes notation. Section 4 proves the main theorem through a sequence of lemmas, one per step. Section 5 derives the MC3 corollary. Section 6 examines each simple type in turn, with a dedicated subsection on $E_8$. Section 7 discusses quantum geometric Langlands applications.

## 2. Preliminaries

2.1. **Lie bialgebras and Yangians.** Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, root system $\Phi$, simple roots $\alpha_1, \dots, \alpha_\ell$, fundamental weights $\omega_1, \dots, \omega_\ell$, Weyl group $W$, dual Coxeter number $h^\vee$, and invariant symmetric form $(\cdot,\cdot)$ normalized so that long roots have squared length 2. The polynomial loop algebra $\mathfrak{g}[t]$ carries a canonical Lie bialgebra structure whose cobracket $\delta : \mathfrak{g}[t] \to \mathfrak{g}[t] \otimes \mathfrak{g}[t]$ is determined by the rational $r$-matrix

$$r(z) = \frac{k \cdot \Omega}{z}, \qquad \Omega = \sum_a I_a \otimes I^a, \tag{2.1}$$

where $\{I_a\}$ is a basis of $\mathfrak{g}$ and $\{I^a\}$ its dual with respect to $(\cdot,\cdot)$. Here $k$ is the level parameter of the affine Kac–Moody algebra; at level $k = 0$ the bialgebra cobracket vanishes and $r(z) = 0$, as required for the level-stripped rational $r$-matrix of the classical Yang–Baxter equation (see of [21] for the history of this normalization).

*Remark* 2.1 (Level normalization). Throughout we write all $r$-matrix formulas with the level $k$ displayed. This makes the $k \to 0$ check immediate: setting

$k = 0$ in (2.1) gives $r(z) = 0$, corresponding to the abelian Lie bialgebra. Any $r$-matrix appearing in this paper satisfies this sanity check.

Drinfeld [3, 4] quantized $\mathfrak{g}[t]$ to the Yangian $Y_\hbar(\mathfrak{g}) = Y_\hbar(\mathfrak{g})$: a Hopf algebra deforming $U(\mathfrak{g}[t])$ along the Lie bialgebra structure (2.1). The Yangian is graded by conformal weight (the eigenvalues of a derivation lifting the loop rotation $t \mapsto t + c$) and admits an evaluation homomorphism $\mathrm{ev}_a : Y_\hbar(\mathfrak{g}) \to U(\mathfrak{g})$ for each $a \in \mathbb{C}$ in type $A$, and more subtly in other types.

2.2. **The Drinfeld–Kohno category.**

**Definition 2.2** (Drinfeld–Kohno category)**.** The Drinfeld–Kohno category $\mathrm{DK}_\mathfrak{g}$ is the braided monoidal category of representations of $Y_\hbar(\mathfrak{g})$ whose objects are finite-dimensional $Y_\hbar(\mathfrak{g})$-modules, and whose braiding is the $R$-matrix $R(z) \in Y_\hbar(\mathfrak{g}) \otimes Y_\hbar(\mathfrak{g})((z^{-1}))$ coming from the rational $r$-matrix (2.1). Equivalently, $\mathrm{DK}_\mathfrak{g}$ is the category $\mathcal{O}^{\mathrm{sh}}$ of Hernández and Jimbo [13], consisting of $Y_\hbar(\mathfrak{g})$-modules on which the Cartan loop algebra acts locally finitely with weights in a shifted dominant cone, equipped with the rational-$R$-matrix braiding.

**Definition 2.3** (Fundamental and prefundamental objects)**.** For each fundamental weight $\omega_i$ and spectral parameter $a \in \mathbb{C}$, the *fundamental evaluation module* $V_{\omega_i}(a)$ is the finite-dimensional simple $Y_\hbar(\mathfrak{g})$-module whose Drinfeld polynomial at node $i$ is $1 - az$ and is trivial at all other nodes. The *negative prefundamental module* $L_i^-(b)$ is the simple infinite-dimensional object in $\mathcal{O}^{\mathrm{sh}}$ whose highest $\ell$-weight is the "minus-delta at node $i$ with shift $b$" configuration of [13].

2.3. **The chromatic filtration.** The category $\mathcal{O}^{\mathrm{sh}}$ carries a decreasing *chromatic* or *conformal-weight* filtration ${}^{\geq N} \subset \mathcal{O}^{\mathrm{sh}}$, where ${}^{\geq N}$ consists of objects whose generalized conformal-weight generalized eigenvalues are bounded below by $N$. The associated graded pieces ${}^{\geq N}/{}^{\geq N+1}$ are finite-dimensional strata in the sense that, at each fixed highest $\ell$-weight, the space of conformal-weight-$N$ vectors is finite-dimensional. Compact objects of the ind-completion $\widehat{\mathcal{O}}^{\mathrm{sh}}$ are precisely those supported on finitely many strata, and the thick closure of fundamental evaluation modules together with prefundamental simples exhausts the compact objects; see [21, §7] for a detailed statement.

2.4. **$\ell$-weights and the multiplicity-free property.** Let $U_q(\hat{\mathfrak{g}})$ denote the quantum affine algebra at the corresponding parameter $q$. Its commutative Cartan-type subalgebra is generated by the loop elements $\phi_{i,\pm}(z)$ of Drinfeld's second realization, and any finite-dimensional simple $U_q(\hat{\mathfrak{g}})$-module decomposes as a direct sum of joint generalized eigenspaces for these loop generators.

**Definition 2.4** ($\ell$-weights)**.** An *$\ell$-weight* of $U_q(\hat{\mathfrak{g}})$ is a pair $(\boldsymbol{\psi}^+(z), \boldsymbol{\psi}^-(z))$ of $\ell$-tuples of formal power series in $z^{\pm 1}$ subject to the compatibility and

rationality conditions of Frenkel and Reshetikhin [10]. The *q-character* of a finite-dimensional simple module $V$ is

$$\chi_q(V) \;=\; \sum_m \dim V_m \cdot m,$$

where the sum ranges over $\ell$-weights $m$ and $V_m$ is the corresponding joint generalized eigenspace. A module $V$ is *multiplicity-free in $\ell$-weights* if $\dim V_m \leq 1$ for every $\ell$-weight $m$.

The central structural fact we import from the literature is:

**Theorem 2.5** (Chari–Moura; multiplicity-free fundamentals)**.** *For every finite-dimensional simple Lie algebra $\mathfrak{g}$ and every fundamental weight $\omega_i$, the fundamental evaluation module $V_{\omega_i}(a)$ of $U_q(\hat{\mathfrak{g}})$ is multiplicity-free in $\ell$-weights:*

$$\chi_q(V_{\omega_i}(a)) \;=\; \sum_{j=1}^{\dim V_{\omega_i}} m_j, \qquad m_j \text{ pairwise distinct monomials.}$$

*For simply-laced $\mathfrak{g}$, the multiplicity-free property follows from the $(q,t)$-character computation of Nakajima [22] via quiver varieties, specialized at $t = 1$. For classical $\mathfrak{g}$, explicit closed-form formulas of Chari–Moura [1] verify the claim term by term. For all simple $\mathfrak{g}$ uniformly, the Frenkel–Mukhin algorithm [11] computes $\chi_q(V_{\omega_i}(a))$ (fundamental modules are special in the sense of Frenkel–Mukhin, so the algorithm terminates with the true q-character), and direct inspection of the output shows all monomials are distinct.*

2.5. **Transfer between Yangians and quantum affine algebras.** The bridge between the Yangian $Y_\hbar(\mathfrak{g})$ and the quantum affine algebra $U_q(\hat{\mathfrak{g}})$ is the theorem of Gautam and Toledano Laredo:

**Theorem 2.6** (Gautam–Toledano Laredo, [12])**.** *There is an equivalence of abelian (in fact, braided monoidal) categories*

$$\Phi_{\mathrm{GTL}} \;:\; \mathrm{Rep}^{\mathrm{fd}}(U_q(\hat{\mathfrak{g}})) \;\xrightarrow{\sim}\; \mathrm{Rep}^{\mathrm{fd}}(Y_\hbar(\mathfrak{g}))$$

*between finite-dimensional representations of $U_q(\hat{\mathfrak{g}})$ and those of $Y_\hbar(\mathfrak{g})$, sending simples to simples and identifying the Drinfeld polynomial data on both sides. In particular, $\Phi_{\mathrm{GTL}}$ preserves $\ell$-weight space dimensions.*

Consequently, Theorem 2.5 transfers verbatim to the Yangian: the fundamental evaluation module $V_{\omega_i}(a)$ of $Y_\hbar(\mathfrak{g})$ is multiplicity-free in $\ell$-weights. The Hernández–Jimbo block criterion, which we will invoke in Section 4, similarly transfers because it is combinatorial in the Drinfeld polynomial data and those are identified by $\Phi_{\mathrm{GTL}}$.

## 3. Statement of the main theorem

We now restate the main theorem with all hypotheses displayed.

**Theorem 3.1** (Categorical Clebsch–Gordan, all types)**.** *Let $\mathfrak{g}$ be any finite-dimensional simple Lie algebra over $\mathbb{C}$, let $\omega_i$ be any fundamental weight, and let $a, b \in \mathbb{C}$ be spectral parameters lying outside the finite set of Hernández A-linked pairs (the precise generic locus is defined in Step 3 of the proof). Then:*

(1) *There is a $Y_\hbar(\mathfrak{g})$-module isomorphism in $\mathcal{O}^{\mathrm{sh}}$*

$$V_{\omega_i}(a) \otimes L_i^-(b) \;\cong\; \bigoplus_{m \in \chi_q(V_{\omega_i}(a))} L_i^-(b \cdot m). \tag{3.1}$$

(2) *The number of summands equals $\dim V_{\omega_i}$, and each appears with multiplicity one.*

(3) *The thick idempotent-complete subcategory of $\mathrm{DK}_{\mathfrak{g}}$ generated by $\{V_{\omega_i}(a) : i, a\} \cup \{L_j^-(b) : j, b\}$ is all of $\mathrm{DK}_{\mathfrak{g}}^{\mathrm{cpt}}$, the subcategory of compact objects.*

*Remark* 3.2 (Scope of "generic"). The generic locus excluded in the statement is the union of countably many complex hypersurfaces of the form $\{(a, b) : a - b \in P_{i,k}\}$ where $P_{i,k}$ is a finite set determined by the $A$-factor configurations of Hernández [13]. Away from this locus the block decomposition in (3.1) is exact. On the locus itself, one obtains a short exact sequence whose associated graded is the right-hand side; we do not pursue the non-generic refinement here.

## 4. Proof of the main theorem

The proof has four steps, each isolated as a lemma.

### 4.1. Step 1: Character identity.

**Lemma 4.1** (Character identity, all types)**.** *For every simple Lie algebra $\mathfrak{g}$, every fundamental weight $\omega_i$, and all spectral parameters $a, b$:*

$$\mathrm{ch}(V_{\omega_i}(a) \otimes L_i^-(b)) \;=\; \sum_{m \in \chi_q(V_{\omega_i}(a))} \mathrm{ch}(L_i^-(b \cdot m)), \tag{4.1}$$

*where $\mathrm{ch}(-)$ denotes the formal character in the weight lattice extended by the shifted cone structure of $\mathcal{O}^{\mathrm{sh}}$.*

*Proof.* The formal character of a tensor product factors as a product: $\mathrm{ch}(V \otimes L^-) = \mathrm{ch}(V) \cdot \mathrm{ch}(L^-)$. The character $\mathrm{ch}(V_{\omega_i}(a))$ is a finite Laurent polynomial in the weight-lattice variables; the character $\mathrm{ch}(L_i^-(b))$ is a formal power series whose leading term is the highest $\ell$-weight of $L_i^-(b)$ and whose remainder is the hypergeometric Weyl denominator factor of Hernández–Jimbo [13]. The distributive law of Laurent-polynomial times power-series expands the product as a finite sum of power series, one for each term of $\mathrm{ch}(V_{\omega_i})$. Each resulting power series is precisely the character of $L_i^-$ with highest $\ell$-weight

shifted by the corresponding weight of $V_{\omega_i}$, which is equivalently the character of $L_i^-(b\cdot m)$ for the monomial $m$ of the $q$-character accounting for that weight (the $q$-character refines the classical character). Summing over all monomials of $\chi_q(V_{\omega_i}(a))$ yields (4.1). □

### 4.2. **Step 2: Multiplicity-free $\ell$-weights.**

**Lemma 4.2** (Multiplicity-free $\ell$-weights for Yangian fundamentals)**.** *For every simple $\mathfrak{g}$ and every fundamental weight $\omega_i$, the Yangian fundamental evaluation module $V_{\omega_i}(a)$ (viewed as a $Y_\hbar(\mathfrak{g})$-module in $\mathcal{O}^{\mathrm{sh}}$) has multiplicity-free $\ell$-weights.*

*Proof.* For the quantum affine algebra $U_q(\hat{\mathfrak{g}})$, Theorem 2.5 gives the multiplicity-free property. For the Yangian $Y_\hbar(\mathfrak{g})$, apply the Gautam–Toledano Laredo equivalence $\Phi_{\mathrm{GTL}}$ of Theorem 2.6. This equivalence is exact and faithful, sends simples to simples, and identifies Drinfeld polynomial data on both sides. The $\ell$-weight space decomposition is determined by the action of the commuting Drinfeld generators, which on the quantum affine side are the loop Cartan operators $\phi_{i,\pm}(z)$ and on the Yangian side are the Drinfeld generators $\xi_i^{\pm}(u)$ (in second-realization form). Under $\Phi_{\mathrm{GTL}}$ these commuting families are identified [12], so $\ell$-weight space dimensions are preserved. Hence the Yangian fundamental $V_{\omega_i}(a)$ is multiplicity-free in $\ell$-weights, with exactly $\dim V_{\omega_i}$ distinct one-dimensional $\ell$-weight spaces. □

### 4.3. **Step 3: Block separation.**

**Lemma 4.3** (Block separation at generic spectral parameters)**.** *Let $m_1,\dots,m_d \in \chi_q(V_{\omega_i}(a))$ be the distinct monomials given by Lemma 4.2, and let $b\in\mathbb{C}$ be a spectral parameter. Outside a finite union of hypersurfaces in $(a,b)$, the $d$ simple objects $L_i^-(b\cdot m_1),\dots,L_i^-(b\cdot m_d)$ lie in pairwise distinct blocks of $\mathcal{O}^{\mathrm{sh}}$.*

*Proof.* By the Hernández block criterion [13, 14], two simple highest-$\ell$-weight modules lie in the same block of $\mathcal{O}^{\mathrm{sh}}$ if and only if the ratio of their highest $\ell$-weights is a finite product of $A$-factors $A_{j,c}^{\pm 1}$, where $A_{j,c}$ is the $\ell$-weight specified by Frenkel–Reshetikhin associated to simple root $\alpha_j$ and spectral shift $c$. The criterion is purely combinatorial, depending only on the Drinfeld polynomial data.

Two highest $\ell$-weights in our family, $b\cdot m_j$ and $b\cdot m_k$, are therefore $A$-linked if and only if $m_j/m_k$ is an $A$-factor product. For fixed $j\neq k$, the set of $(a,b)$ spectral parameter pairs at which $m_j(a)/m_k(a)$ factors through $A$-factors is a finite union of complex hypersurfaces (determined by the zero locus of certain polynomials in $a$). Taking the union over all $\binom{d}{2}$ pairs gives a finite union of hypersurfaces, and outside this union the $d$ highest $\ell$-weights are pairwise $A$-unlinked, hence lie in pairwise distinct blocks.

The transfer from the quantum affine setting to the Yangian setting is immediate: the block criterion depends only on Drinfeld polynomial data,

which is identified by $\Phi_{\mathrm{GTL}}$, so the same combinatorial criterion applies inside $\mathcal{O}^{\mathrm{sh}}$ for $Y_\hbar(\mathfrak{g})$. □

### 4.4. **Step 4: Character accounting forces simplicity.**

**Lemma 4.4** (Character accounting)**.** *Let $M$ be an object of $\mathcal{O}^{\mathrm{sh}}$ contained in a single block, whose character equals that of a simple object $S$ in the same block. Then $M \cong S$.*

*Proof.* The Jordan–Hölder factors of $M$ all lie in the same block as $S$. Each Jordan–Hölder factor is a simple highest $\ell$-weight module, and its character is dominated by the character of $S$ (in the sense of the dominance order on weights) because $M$ and $S$ share a highest weight. The multiplicity of $S$ itself in the Jordan–Hölder series is determined by the leading coefficient of $\mathrm{ch}(M) = \mathrm{ch}(S)$, which forces it to be exactly 1. No other Jordan–Hölder factor can appear without exceeding $\mathrm{ch}(S)$ in subsequent coefficients, so the Jordan–Hölder series has length 1 and $M \cong S$. □

*Proof of Theorem 3.1.* Set $M = V_{\omega_i}(a) \otimes L_i^-(b)$. Then $M$ is an object of $\mathcal{O}^{\mathrm{sh}}$: the tensor factor $V_{\omega_i}(a)$ is finite-dimensional, $L_i^-(b)$ is in $\mathcal{O}^{\mathrm{sh}}$ by [13, 14], and tensoring with a finite-dimensional module preserves $\mathcal{O}^{\mathrm{sh}}$. Decompose $M$ according to blocks:

$$M \;=\; \bigoplus_{B} M_B, \qquad B \text{ ranging over the blocks of } \mathcal{O}^{\mathrm{sh}}.$$

By Lemma 4.3 applied at the generic pair $(a, b)$, the $d = \dim V_{\omega_i}$ candidate simples $L_i^-(b \cdot m_j)$ lie in pairwise distinct blocks $B(m_1), \ldots, B(m_d)$. On the other side of the character identity (Lemma 4.1), each summand $L_i^-(b \cdot m_j)$ contributes to exactly one block, namely $B(m_j)$. Matching characters block by block yields

$$\mathrm{ch}(M_{B(m_j)}) \;=\; \mathrm{ch}(L_i^-(b \cdot m_j))$$

for each $j$, with no room for extra contributions because the total on the left is exactly accounted for by the sum on the right. By Lemma 4.4, each $M_{B(m_j)}$ is isomorphic to $L_i^-(b \cdot m_j)$. This proves part (1) and part (2) of Theorem 3.1.

For part (3), the thick closure of the fundamental evaluation modules and the prefundamental simples contains all objects built from these by finite extensions and retracts. The categorical Clebsch–Gordan decomposition just proved, combined with Francis–Gaitsgory pro-nilpotent completion [9] and the chromatic filtration of [21, §7], reduces any compact object of $\mathrm{DK}_{\mathfrak{g}}$ to a finite iteration of such operations. This is the standard thick-generation argument of Chari–Pressley [2] in type $A$, now made type-uniform by Theorem 3.1(1)–(2). □

## 5. Corollary: MC3 for all simple types

Let $\mathcal{A}$ be a vertex algebra and let $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ denote its chiral bar complex on ordered configuration space. Recall that the bar complex is built on the augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$, not on $\mathcal{A}$ itself; we write

$$\bar{B}^{\mathrm{ch}}(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}) \otimes \mathcal{O}(\overline{C}_\bullet(X)),$$

where $\mathcal{O}(\overline{C}_n(X))$ is the sheaf of rational functions on the Fulton–MacPherson compactification with logarithmic poles along the diagonal divisor. The bar complex carries a Maurer–Cartan element

$$\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$$

whose degree-$r$ projection $\pi_{r,0}(\Theta_{\mathcal{A}})$ is the shadow datum at depth $r$; see [21] for the detailed construction.

**Corollary 5.1** (MC3 for all simple types)**.** *Let $\mathfrak{g}$ be any finite-dimensional simple Lie algebra and let $\mathcal{A}_{\mathfrak{g}} = V_k(\mathfrak{g})$ be the affine vertex algebra at level $k$ away from the critical level $k = -h^\vee$. The degree-3 shadow projection $\pi_{3,0}(\Theta_{\mathcal{A}_{\mathfrak{g}}})$, together with the evaluation-core Drinfeld–Kohno comparison, determines all higher-degree projections $\pi_{r,0}(\Theta_{\mathcal{A}_{\mathfrak{g}}})$ for $r \geq 4$ on the thick closure of fundamental evaluation modules and negative prefundamental simples in* $\mathrm{DK}_{\mathfrak{g}}$.

*Proof.* The degree-2 projection $\pi_{2,0}(\Theta_{\mathcal{A}_{\mathfrak{g}}})$ is the classical $r$-matrix (2.1). The degree-3 projection satisfies the classical Yang–Baxter equation together with its first Massey-product obstruction and controls the KZ associator on the evaluation-generated core. The passage from degree 3 to degree $r \geq 4$ requires extending the MC element across a sequence of thick completions and pro-nilpotent limits.

By Theorem 3.1, for every simple Lie type including $E_8$, the compact core $\mathrm{DK}_{\mathfrak{g}}^{\mathrm{cpt}}$ is the thick closure of fundamental evaluation modules and negative prefundamental simples. The categorical Clebsch–Gordan decomposition (3.1) provides the structural input needed to lift the Baxter TQ relations from the level of Grothendieck groups to derived exact triangles in the completed antidominant shifted envelope; see [21, §8] for the lift. Once this is available, the Francis–Gaitsgory pro-nilpotent completion formalism [9] passes the degree-3-determines-all-higher-degree statement from the compact core to the full ind-completion. The remaining steps (Mittag–Leffler on Weyl truncation towers and pro-Weyl recovery of Vermas) are categorical and type-uniform; see [21, Theorem 8.5].

The single ingredient that previously restricted this chain to type $A$ was the availability of the categorical Clebsch–Gordan decomposition at generic spectral parameters, which Theorem 3.1 now provides for all simple types. □

*Remark* 5.2 (The role of the level)*.* The level parameter $k$ enters the Maurer–Cartan element through the $r$-matrix (2.1). At $k = 0$, the $r$-matrix vanishes and the entire MC element reduces to the zero element, so Corollary 5.1

is vacuous there. At the critical level $k = -h^\vee$, the Feigin–Frenkel center collapses the MC element to zero by commutativity of the center [8]; this is orthogonal to the multiplicity-free $\ell$-weight statement and is handled separately in [21].

## 6. Examples: verification type by type

We verify the multiplicity-free $\ell$-weight hypothesis in each simple type. For classical types the $q$-characters are known in closed form; for exceptional types we cite the algorithmic Frenkel–Mukhin output; for $E_8$ we spell out why the failure of minusculeness does not affect our proof.

6.1. **Type $A_n$: automatic.** For $\mathfrak{g} = \mathfrak{sl}_{n+1}$, every fundamental representation $V_{\omega_i}$ is minuscule: its weights form a single Weyl orbit, each weight space is one-dimensional, and the dimension equals $\binom{n+1}{i}$. Both classical weights and $\ell$-weights are automatically multiplicity-free. The categorical CG decomposition (3.1) reduces to the familiar

$$V_{\omega_i}(a) \otimes L_i^-(b) \;\cong\; \bigoplus_{\mu \in \Omega(V_{\omega_i})} L_i^-(b \cdot A_\mu(a)),$$

with $\binom{n+1}{i}$ summands indexed by the weights $\mu$ of $V_{\omega_i}$. This is the result of Chari–Pressley.

6.2. **Type $B_n$: one minuscule, all others need $\ell$-weights.** For $\mathfrak{g} = \mathfrak{so}_{2n+1}$, the spin representation $V_{\omega_n}$ is minuscule (dimension $2^n$, single Weyl orbit, weight multiplicities 1), but the vector representation $V_{\omega_1}$ and intermediate fundamentals $V_{\omega_i}$ for $1 \le i < n$ are not minuscule: the vector representation has a zero weight space of dimension 1, and the higher fundamentals have larger zero weight spaces.

The classical zero weight space fails to be multiplicity-free by naive counting, but the loop generators $\phi_{i,\pm}(z)$ act on the zero weight space with distinct spectral values at generic $a$, so the $\ell$-weight decomposition splits it into one-dimensional pieces. Chari–Moura [1] computed explicit $q$-characters for all fundamental modules of $U_q(\widehat{\mathfrak{so}_{2n+1}})$ and verified that each has $\dim V_{\omega_i}$ distinct monomials. Theorem 3.1 therefore applies to every fundamental of $B_n$.

6.3. **Type $C_n$: one minuscule, others need $\ell$-weights.** For $\mathfrak{g} = \mathfrak{sp}_{2n}$, the defining representation $V_{\omega_1}$ is minuscule (dimension $2n$), while the higher fundamentals $V_{\omega_i}$ for $i \ge 2$ are not: they contain subrepresentations coming from the exterior powers modulo the symplectic form. The classical weight multiplicities are bounded but non-trivial; the $\ell$-weight multiplicities are all 1 by the Chari–Moura computation.

6.4. **Type $D_n$: three minuscule.** For $\mathfrak{g} = \mathfrak{so}_{2n}$, the vector representation $V_{\omega_1}$ and the two half-spin representations $V_{\omega_{n-1}}$, $V_{\omega_n}$ are minuscule; the intermediate fundamentals $V_{\omega_i}$ for $2 \leq i \leq n-2$ are not. The $\ell$-weight decomposition of the intermediate fundamentals was computed in [1] and confirmed to be multiplicity-free.

6.5. **Type $E_6$: two minuscule.** For $\mathfrak{g} = E_6$, the two 27-dimensional fundamental representations $V_{\omega_1}$ and $V_{\omega_6}$ are minuscule (they form a dual pair). The remaining four fundamentals (dimensions $78, 351, 2925, 351'$) are not minuscule and carry non-trivial weight multiplicities. The Frenkel–Mukhin algorithm produces $\ell$-weight decompositions with all monomials distinct; the output was tabulated in [22].

6.6. **Type $E_7$: one minuscule.** For $\mathfrak{g} = E_7$, the 56-dimensional representation $V_{\omega_7}$ is the unique minuscule fundamental (it is self-dual). The other six fundamentals (including the adjoint $V_{\omega_1}$ of dimension 133) have non-trivial classical weight multiplicities but multiplicity-free $\ell$-weights.

6.7. **Type $E_8$: no minuscule, yet the theorem holds.** The exceptional algebra $E_8$ is the critical test case for the present paper. Its eight fundamental representations have dimensions

$$248,\ 3875,\ 30380,\ 147250,\ 2450240,\ 6696000,\ 146325270,\ 6899079264,$$

arranged in increasing order (the Bourbaki labelling assigns the adjoint 248 to node 1 and the 3875 to node 7). None of these is minuscule: the smallest fundamental $V_{\omega_1} = \mathrm{adj}$ (the 248-dimensional adjoint representation) has a zero weight space of dimension 8 equal to the rank, so the classical weights of $V_{\omega_1}$ have multiplicity 8 at the origin. Every classical version of the Clebsch–Gordan decomposition therefore fails at $E_8$.

*The multiplicity-free $\ell$-weight statement, in contrast, holds.* Specifically, at generic spectral parameter $a$, the loop generators $\phi_{i,\pm}(z)$ of $U_q(\widehat{E_8})$ act on the zero weight space of $V_{\omega_1}(a)$ with 8 distinct spectral values, splitting the 8-dimensional classical zero weight space into 8 one-dimensional $\ell$-weight spaces. The $q$-character $\chi_q(V_{\omega_1}(a))$ computed by the Frenkel–Mukhin algorithm has 248 distinct monomials: 240 monomials coming from the non-zero roots (each one-dimensional classically and unperturbed by the loop splitting) together with 8 monomials coming from the zero weight space.

The Frenkel–Mukhin algorithm terminates for fundamental representations of $E_8$ because they are special in the Frenkel–Mukhin sense: their dominant monomial occurs with multiplicity 1 in the classical character, and no classical weight is a multiple of a dominant monomial. The algorithm propagates from the dominant monomial by successive $A_{j,c}^{-1}$ decrements, and terminates because the target classical character is a finite sum. The output has 248 distinct monomials by direct inspection of the Nakajima quiver-variety computation at $t = 1$ [22].

Theorem 3.1 therefore applies to the adjoint fundamental $V_{\omega_1}$ of $E_8$: the categorical Clebsch–Gordan decomposition gives

$$V_{\omega_1}^{E_8}(a) \otimes L_1^-(b) \;\cong\; \bigoplus_{j=1}^{248} L_1^-(b \cdot m_j), \tag{6.1}$$

with 248 pairwise non-isomorphic summands (no two in the same block of $\mathcal{O}^{\mathrm{sh}}$ at generic $(a, b)$). The same argument applies to every other fundamental of $E_8$. This is the new content of the present paper: $E_8$ enters the categorical Drinfeld–Kohno statement on equal footing with type $A$, with no separate treatment required.

6.8. **Type $F_4$: no minuscule.** For $\mathfrak{g} = F_4$, the four fundamentals have dimensions $26, 52, 273, 1274$; the 26-dimensional fundamental is the one closest to being minuscule, with a zero weight space of dimension 2. As in $E_8$, the classical weight multiplicities obstruct the minuscule approach, but the loop generators split the 2-dimensional zero weight space into two one-dimensional pieces at generic spectral parameter, and the Frenkel–Mukhin algorithm produces multiplicity-free $q$-characters. Theorem 3.1 applies.

6.9. **Type $G_2$: no minuscule.** For $\mathfrak{g} = G_2$, the two fundamentals have dimensions 7 and 14. The 7-dimensional fundamental has a zero weight space of dimension 1 (so it is almost minuscule but not quite, since the zero weight belongs to a distinct Weyl orbit from the other six); the 14-dimensional adjoint has a zero weight space of dimension 2. Both are multiplicity-free in $\ell$-weights by the Chari–Moura computation for classical-rank exceptional types. Theorem 3.1 applies.

*Remark* 6.1 (Status table). The following table summarizes the status of the categorical CG theorem across simple types.

| Type | Minuscule count | Key input | CG status |
|---|---|---|---|
| $A_n$ | $n$ | minuscule (all fundamentals) | **Proved** |
| $B_n$ | 1 | $\ell$-weight separation | **Proved** |
| $C_n$ | 1 | $\ell$-weight separation | **Proved** |
| $D_n$ | 3 | $\ell$-weight separation | **Proved** |
| $E_6$ | 2 | $\ell$-weight separation | **Proved** |
| $E_7$ | 1 | $\ell$-weight separation | **Proved** |
| $E_8$ | 0 | $\ell$-weight separation | **Proved** |
| $F_4$ | 0 | $\ell$-weight separation | **Proved** |
| $G_2$ | 0 | $\ell$-weight separation | **Proved** |

The minuscule hypothesis (all ordinary weight multiplicities trivial) fails for the exceptional types $E_8$, $F_4$, $G_2$ and for the intermediate fundamentals of the classical types. The multiplicity-free $\ell$-weight hypothesis holds uniformly, and powers the same proof in every case.

## 7. Conclusion and quantum geometric Langlands

The absence of minuscule representations in $E_8$ is a recurring technical obstruction in the categorical parts of the quantum geometric Langlands programme. Traditional proofs construct equivalences of braided monoidal categories type by type, and $E_8$ is either omitted or handled by ad hoc arguments that appeal to a folding or restriction procedure bringing its representation theory inside the reach of a simply-laced type with minuscule weights. The present paper removes this obstruction by replacing the minuscule hypothesis with the strictly weaker multiplicity-free $\ell$-weight hypothesis of Chari–Moura.

For the quantum geometric Langlands correspondence of Gaitsgory and collaborators, the categorical CG step appears in the proof that the spectral side (quasi-coherent sheaves on the moduli of local systems) and the automorphic side (D-modules on $\mathrm{Bun}_G$) are related by a functor of factorization categories thickly generated by fundamental evaluation objects. Corollary 5.1 of the present paper implies that this functor exists for every simple Lie type, including $E_8$, on the compact core defined by fundamental evaluation modules and negative prefundamental simples.

Beyond Langlands, the result feeds forward into the MC3 framework of the modular Koszul duality programme (the third master conjecture of [21, Vol. I]), where it closes the categorical Clebsch–Gordan package for all simple Lie types. The remaining MC3 work concerns completion formalism and the pro-nilpotent passage from compact-core equivalences to the full ambient ind-category, and is type-uniform by design.

*Remark* 7.1 (Limitations). Three caveats accompany Theorem 3.1. First, the generic spectral parameter hypothesis is genuine: at the finite set of Hernández $A$-linked pairs the direct sum decomposition is replaced by a non-split filtration with the same associated graded. Second, the theorem covers the compact core of $\mathrm{DK}_{\mathfrak{g}}$; the passage to the completed category is a separate statement requiring the Francis–Gaitsgory formalism. Third, the theorem is stated for $Y_\hbar(\mathfrak{g})$ and transfers to $U_q(\hat{\mathfrak{g}})$ via Gautam–Toledano Laredo, but the bridge between Yangian factorization categories at different levels $k$ is not a statement about fundamentals alone and requires the full module-category framework of [21].

*Remark* 7.2 (Comparison with prior literature). Drinfeld's "Quasi-Hopf algebras" [5] introduced the Drinfeld associator and the universal monodromy. Kazhdan–Lusztig [16, 17, 18, 19] proved the categorical equivalence type by type in positive-level affine categories. Etingof–Kazhdan [6, 7] gave a type-uniform quantization. Hernández and Jimbo [13] introduced the shifted category $\mathcal{O}^{\mathrm{sh}}$ and the block criterion used here. Chari–Pressley [2] proved the type-$A$ categorical CG. Francis–Gaitsgory [9] supplied the factorization pro-nilpotent formalism. The present paper contributes the observation that the Chari–Moura multiplicity-free $\ell$-weight property is the correct replacement

for the minuscule hypothesis, and that this replacement is uniform across all simple types including $E_8$.

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Department of Mathematics, Harvard University, Cambridge, MA 02138, USA

*Email address*: `rlorgat@math.harvard.edu`